

ANTI – DEPRESSANTS

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What is Depression?

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- A heterogeneous disorder of Mood;
- Ranges from very mild condition (bordering on normality) to severe psychotic depression (accompanied by hallucinations and delusions)
- Lifetime prevalence: Approx. 10-12% of population
- A major cause of disability and premature death worldwide

Manifestations of Depression

1. Depression with Typical symptoms:

- Sadness, loss of libido, disturbance in appetite, weight loss, negative emotions and decreased sleep
- Suicidal tendencies and Psychotic symptoms (e.g. *Hallucinations and Delusions in severe depression*)

2. Depression with Atypical symptoms:

- Anxiety, Phobias, Obsessions, Hypochondriasis (*multiple body complaints*), Over-reacting, increased sleep tendency, etc.

Types of Depression

1. **Unipolar depression:**

- Mood swings constantly in same direction (i.e. negative depressive mood)
- *Most common type; 75% non-familial, associated with stressful life events; 25% endogenous, familial*

2. **Bipolar affective disorder:**

- Depression alternating with mania
- *Usually presents in early adult life; strong hereditary tendency*

Etiology theories

1. **Mono-Amine theory of Depression** (Schildkraut, 1965)

- Decrease in the activity of *Norepinephrine* and *5-HT* (brain biogenic amine neurotransmitters responsible for mood) will result in depression
- Whereas, increase in their activity will result in mood elevation.

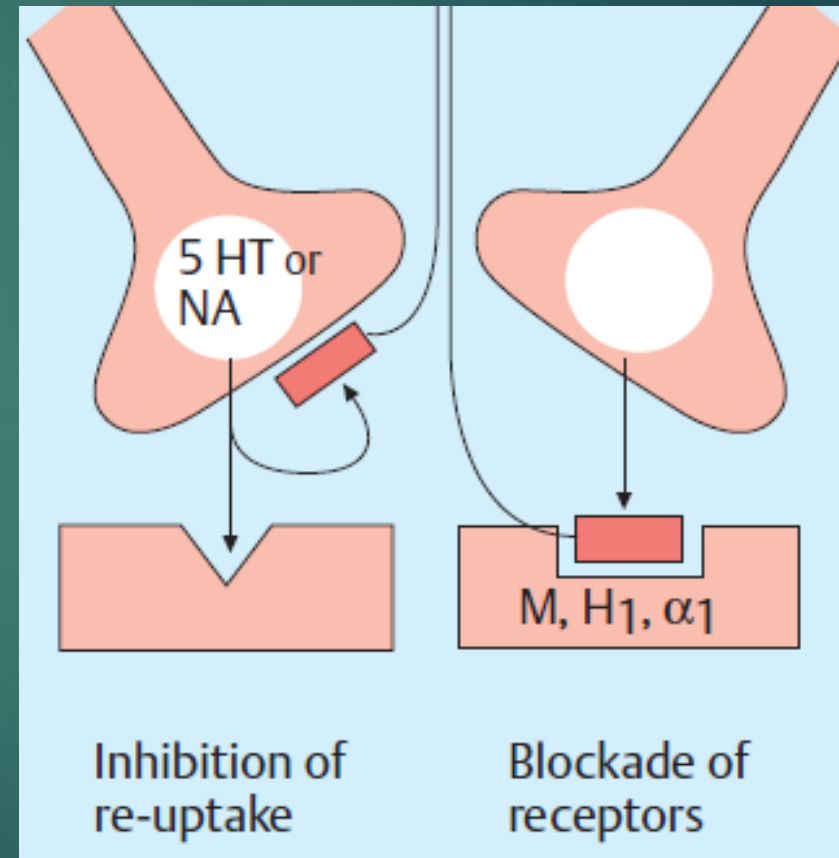
2. **Neuroplasticity & Trophic effects theory** (Duman, Charney et al, 2004)

- Major depression is associated with *neuronal loss* in the hippocampus and prefrontal cortex of brain

Antidepressant Drugs: Mode of Action

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1. Elevation of brain biogenic amines: NE and/or 5-HT
(Immediately within 12 hours);
2. Down regulation of adrenergic and 5-HT₂ receptors in the brain
(Delayed for 2-3 weeks).



Luellmann, 2005

Antidepressant Drugs: Classification

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1. Inhibitors of monoamine uptake inhibitors(Amine pump inhibitors):

- Non-selective (NE & 5-HT) uptake inhibitors, e.g. Tricyclic Antidepressants (*imipramine, amitriptyline, clomipramine etc*), *Venlafaxine, Duloxetine*, etc
- Selective serotonin (5-HT) reuptake inhibitors (SSRIs) (e.g. *fluoxetine, fluvoxamine, paroxetine, dapoxetine & sertraline*)
- Selective noradrenaline (NE) reuptake inhibitors (e.g. *maprotiline, reboxetine*)

2. Mono-amine Oxidase Inhibitors (MAOI)

- *These drugs inhibit MAO type A enzyme, thus inhibiting metabolism of biogenic amine (NE & 5-HT) stores available for release in synaptic cleft*
- Examples include Trancypamine, Moclobemide, Phenelzine

3. Miscellaneous:

Presynaptic adrenergic α_2 receptor blockers

- Blockade of these receptors results in increased release of biogenic amines into the synaptic cleft
- *Examples include Mianserine, Trazodone*

Antidepressant Drugs

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Drug Name	Clinical Uses	Adverse Effects
Tricyclic Antidepressants (TCAs)		
Amitriptyline	Depression, Adjuvant analgesia, Nocturnal enuresis	Sedation, Orthostatic hypotension, Anticholinergic effects (e.g. Dry mouth, etc), Lethergy, Constipation, etc
Imipramine	Depression, nocturnal enuresis	Sedation, Orthostatic hypotension, Anticholinergic effects (e.g. Dry mouth, etc), Lethergy, Constipation, etc
Clomipramine	Obsessive Compulsive Disorder	<i>Similar to Imipramine</i>

Other TCA's: Desipramine, Doxepine, Nortriptyline, Amoxapine, etc

Antidepressant Drugs

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Drug Name	Clinical Uses	Adverse Effects
MAOI's		
Trancypamine	Atypical & Neurotic Depression	<i>Orthostatic hypotension, vertigo, constipation, hypertensive crisis, anticholinergic symptoms, etc</i>
Phenelzine	Neurotic & Atypical Depression	<i>Orthostatic hypotension, vertigo, constipation, hypertensive crisis, anticholinergic symptoms, etc</i>

Other MAOI's: Moclobemide, etc

Antidepressant Drugs

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Drug Name	Clinical Uses	Adverse Effects
SSRI's		
Fluoxetine	Depression, OCD, Bulimia nervosa,	Anxiety, nervousness, insomnia, drowsiness, fatigue, asthenia, tremor, sweating, dizziness, sexual dysfunction, anorexia, etc
Fluvoxamine	Depression, OCD	Headache, insomnia, anxiety, tremor, dizziness, sexual dysfunction, anticholinergic symptoms, etc.
Sertraline	Depression, PTSD, OCD, Panic disorders	Insomnia, nervousness, anxiety, fatigue, nausea, visual disturbance, headache, etc

Other SSRI's: Citalopram, Paroxetine, etc

Antidepressant Drugs

Drug Name	Clinical Uses	Adverse Effects
SNRI's		
Maprotiline	Depression	<i>Hypotension, Sedation, Anticholinergic symptoms, disturbed concentration, etc</i>
Miscellaneous		
Venlafaxine	Depression, Anxiety disorders	<i>Abnormal dreams, Hypertension, weakness, weight loss, parathesia, etc</i>
Trazodone	Depression	<i>Priapism, hypertension, aggression, skin reaction, anaemia, anticholinergic symptoms, etc</i>

Other Antidepressants: Duloxetine, Nefazodone, Mirtazepine, Bupropion, etc

The tyramine (Cheese) effect

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- ▶ Monoamine oxidase inhibitors may lead to accumulation of noradrenaline and serotonin
- ▶ On the other end, some foods like dairy products like cheese and the like are rich in tyramine
- ▶ Tyramine is known to promote the production of catecholamines
- ▶ Taking MAO I's concomitantly with tyramine rich food will lead to over accumulation of catecholamines which among many effects causes hypertensive crisis
- ▶ This effect is known as Tyramine (Cheese) effect
- ▶ It is therefore advised to avoid taking tyramine rich foods when one is taking MAO I's

Contraindications & Interactions

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- TCA's: Hypersensitivity, Avoid concurrent use with MAOI's, Myocardial infarction, Pregnancy & Lactation
- MAOIs + Sympathomimetics → Hypertensive crisis
- MAOIs + TCA → Hypertensive crisis + Hyperthermia + Convulsions
- MAOIs + SSRIs → Serotonin syndrome leading to Hyperthermia and Convulsions
- MAOIs + Opioids → Respiratory depression + Hyperthermia + Convulsions
- Patients on MAOIs should not receive TCA or SSRIs/SNRIs until after 2 weeks from cessation of MAOIs
- Patients on MAOIs should avoid Tyramine-rich foods (e.g. Mature cheese, beer, chicken liver, fermented meats, coffee, milk products, etc) → Catecholamine-induced Hypertensive crisis

END